

AI Credit Risk & Loan Default Prediction Platform

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Abstract

This paper presents an end-to-end AI credit risk and loan default prediction platform designed for banking and retail clients. The system integrates data preprocessing, feature engineering, supervised machine learning models, explainable risk scoring, and a secure dual-portal interface for bank staff and clients. The platform extends traditional borrower assessment by incorporating behavioral indicators, affordability analysis, and live equity holdings to estimate financial resilience. The result is a practical decision-support tool that delivers fast risk insights, transparent recommendations, and portfolio-level analytics.

Keywords

Credit Risk, Loan Default Prediction, Machine Learning, Explainable AI, Streamlit, Risk Scoring, Portfolio Analytics, Market Data Integration.

1. Introduction

Banking institutions require accurate and explainable tools to evaluate loan risk under evolving economic conditions. Conventional rules-based scoring can be opaque or overly static, while modern lending demands real-time analytics, richer feature signals, and user-friendly decision support. The proposed CreditAI platform addresses these needs by combining supervised learning with engineered financial indicators, interactive dashboards, and explainability visualizations.

The paper structure and presentation are inspired by the professional format observed in a recent Random Forest research article, adapted to the credit risk domain. We describe the system architecture, methodology, interface, and results, and then discuss limitations and future scope.

2. System Overview

The platform is implemented as a Streamlit application with two secure portals: (i) Bank Dashboard for applicant evaluation and model analytics, and (ii) Client Portal for eligibility estimation and personal risk insights. The system includes a live Market Watch module to display top stock prices and change percentages, and a stock holdings component that augments borrower assets during scoring.

3. Methodology

The system follows a standard supervised learning pipeline: data ingestion, preprocessing, feature engineering, model training, evaluation, and deployment. Key borrower features include income, loan terms, outstanding obligations, credit history length, payment behavior, and spending indicators. Engineered features capture credit utilization, repayment stability, and loan burden. A scoring engine converts predicted default probability into a credit score range with risk tier classification.

4. Model Training and Evaluation

Three models were evaluated: Logistic Regression, Random Forest, and XGBoost. The best-performing model is selected based on ROC-AUC and overall stability. Performance metrics are persisted and visualized in the Model Analytics dashboard.

4.1 Metrics Summary

Best model: Logistic Regression.

Logistic Regression: Accuracy 0.7396, ROC-AUC 0.7876, Precision 0.1141, Recall 0.6847

Random Forest: Accuracy 0.9354, ROC-AUC 0.7670, Precision 0.2179, Recall 0.1532

XGBoost: Accuracy 0.9504, ROC-AUC 0.7429, Precision 0.1667, Recall 0.0180

5. Client-Side Eligibility and Live Asset Integration

Clients can evaluate how much loan they can afford based on a DTI-based capacity model that compares monthly income against expenses and estimated EMI. The system also accepts equity holdings (ticker and shares) and uses live prices to compute portfolio value. This value is integrated into the applicant assets, enabling a more realistic assessment of repayment capacity.

A Market Watch page displays live prices and changes for major equities, enabling users to see how market movement may affect their asset-based eligibility.

6. Results and Interface Outputs

The following figures demonstrate key system outputs: secure login, bank home dashboard, model analytics, feature importance visualization, applicant evaluation inputs, client portal with stock holdings, market watch, and risk assessment results.

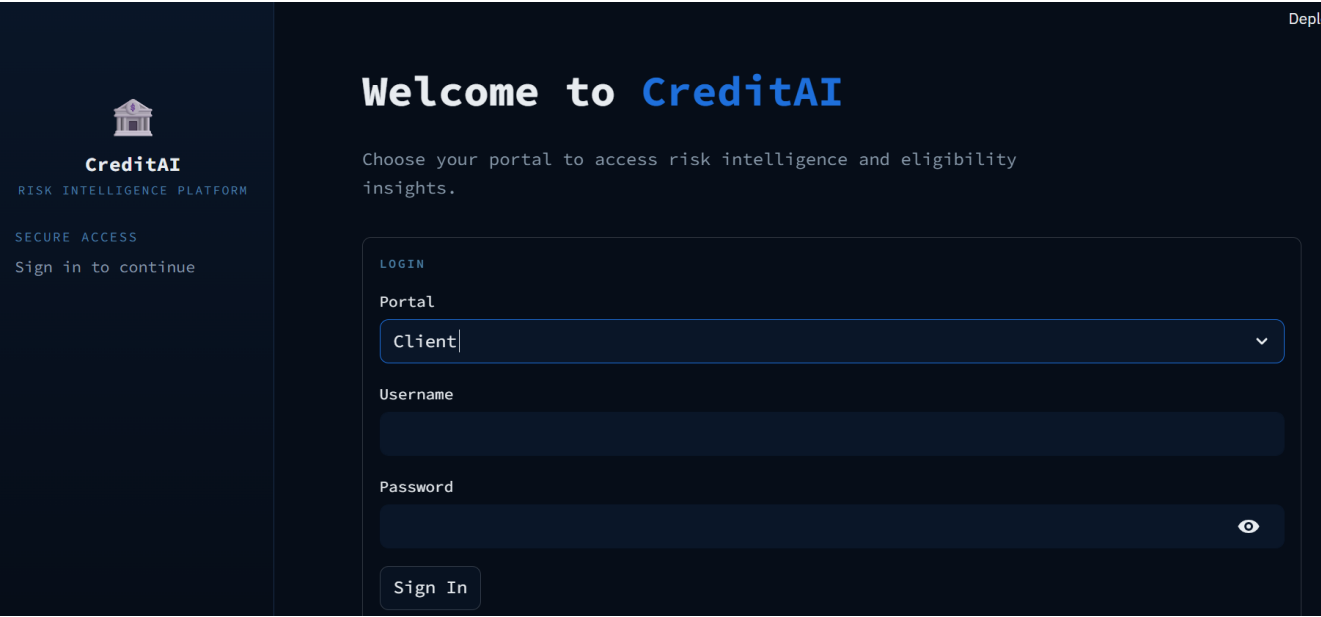


Figure 1: Login page with secure access and portal selection.

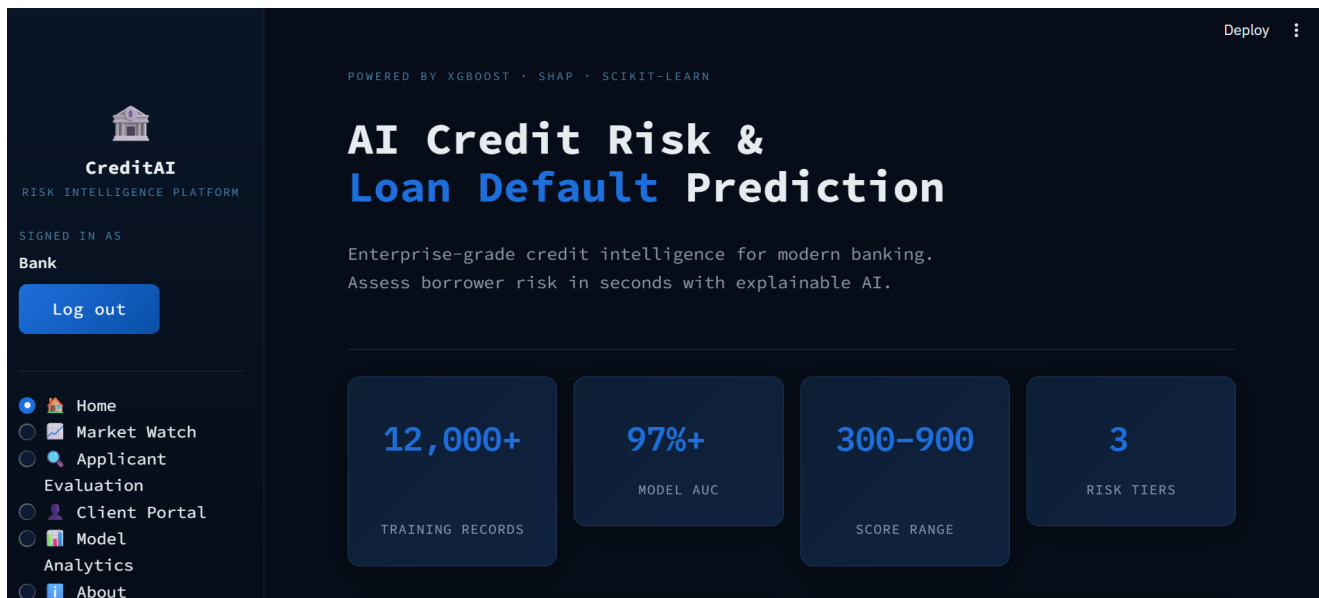


Figure 2: Bank dashboard home with platform highlights.



Figure 3: Model analytics: comparison of candidate models.

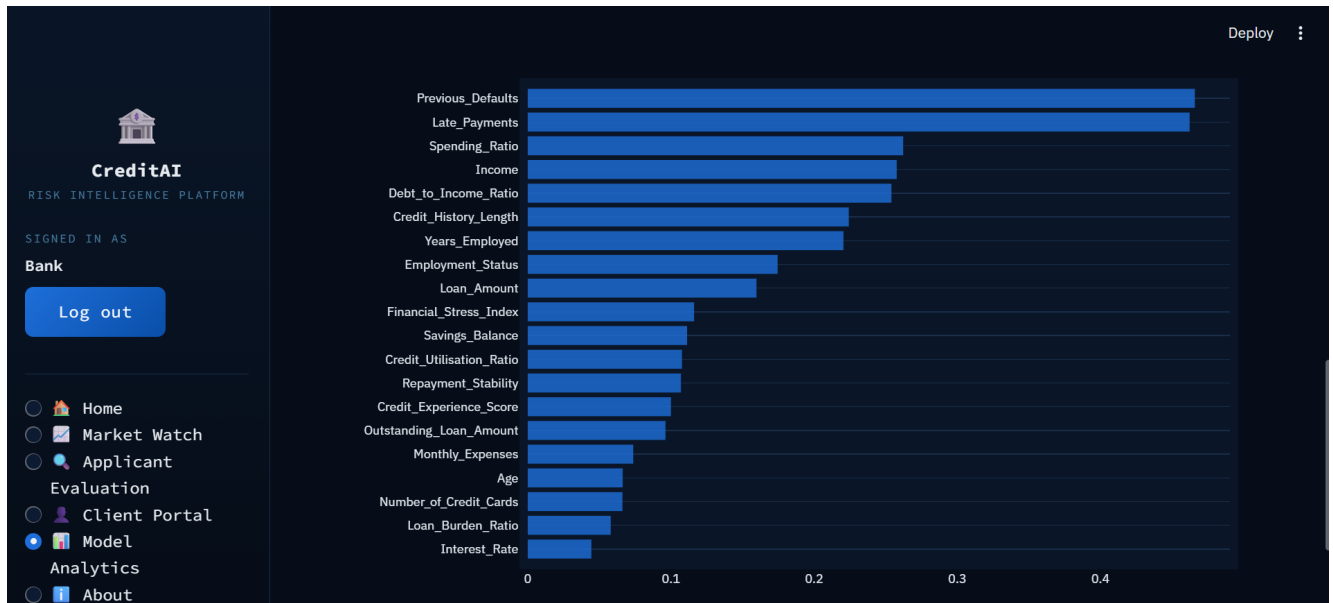


Figure 4: Feature importance for the best-performing model.

Age	Loan Amount (₹)	Credit History (years)
35 - +	300000 - +	10 - +
Annual Income (₹)	Loan Term (months)	Number of Credit Cards
600000 - +	36 v	2 - +
Employment Status	Interest Rate (%)	Savings Balance (₹)
Employed v	10.50 - +	150000 - +
Years Employed	Outstanding Loans (₹)	Monthly Expenses (₹)
8 - +	50000 - +	25000 - +
RISK INDICATORS		
Previous Defaults	Late Payments (count)	Monthly Transactions
0 - +	1 - +	30 - +
Computed Debt-to-Income Ratio: 0.583		
Evaluate Applicant		

Figure 5: Applicant evaluation form for bank users.



CreditAI
RISK INTELLIGENCE PLATFORM

SIGNED IN AS
Client

Log out

- Client Home
- Market Watch
- Client Portal**
- About

MODEL STATUS
ONLINE

Age

32 - +

Desired Loan Amount (₹)

400000 - +

Credit History (years)

8 - +

Annual Income (₹)

650000 - +

Loan Term (months)

36 v

Number of Credit Cards

2 - +

Employment Status

Employed v

Interest Rate (%)

10.50 - +

Savings Balance (₹)

120000 - +

Years Employed

6 - +

Outstanding Loans (₹)

50000 - +

Monthly Expenses (₹)

22000 - +

STOCK HOLDINGS

Stock value is based on live prices and can change.

+ @ ⬇ 🔍 ⌵

Ticker	Shares
GOOGL	100

Figure 6: Client portal with stock holdings input and eligibility form.



CreditAI
RISK INTELLIGENCE PLATFORM

SIGNED IN AS
Client

Log out

- Client Home
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MODEL STATUS
ONLINE

Market Watch

Live prices and changes for major stocks. Data updates every 60 seconds.

	Ticker	Price	Change	Change %
0	AAPL	₹253.51	+3.39	+1.36%
1	MSFT	₹397.15	+1.60	+0.40%
2	GOOGL	₹303.80	+1.52	+0.50%
3	AMZN	₹208.92	+1.25	+0.60%
4	NVDA	₹183.95	+3.70	+2.05%
5	TSLA	₹399.94	+8.74	+2.23%
6	META	₹629.06	+15.35	+2.50%
7	JPM	₹286.90	+3.46	+1.22%
8	V	N/A	N/A	N/A
9	WMT	N/A	N/A	N/A

Figure 7: Market Watch with live prices and changes.

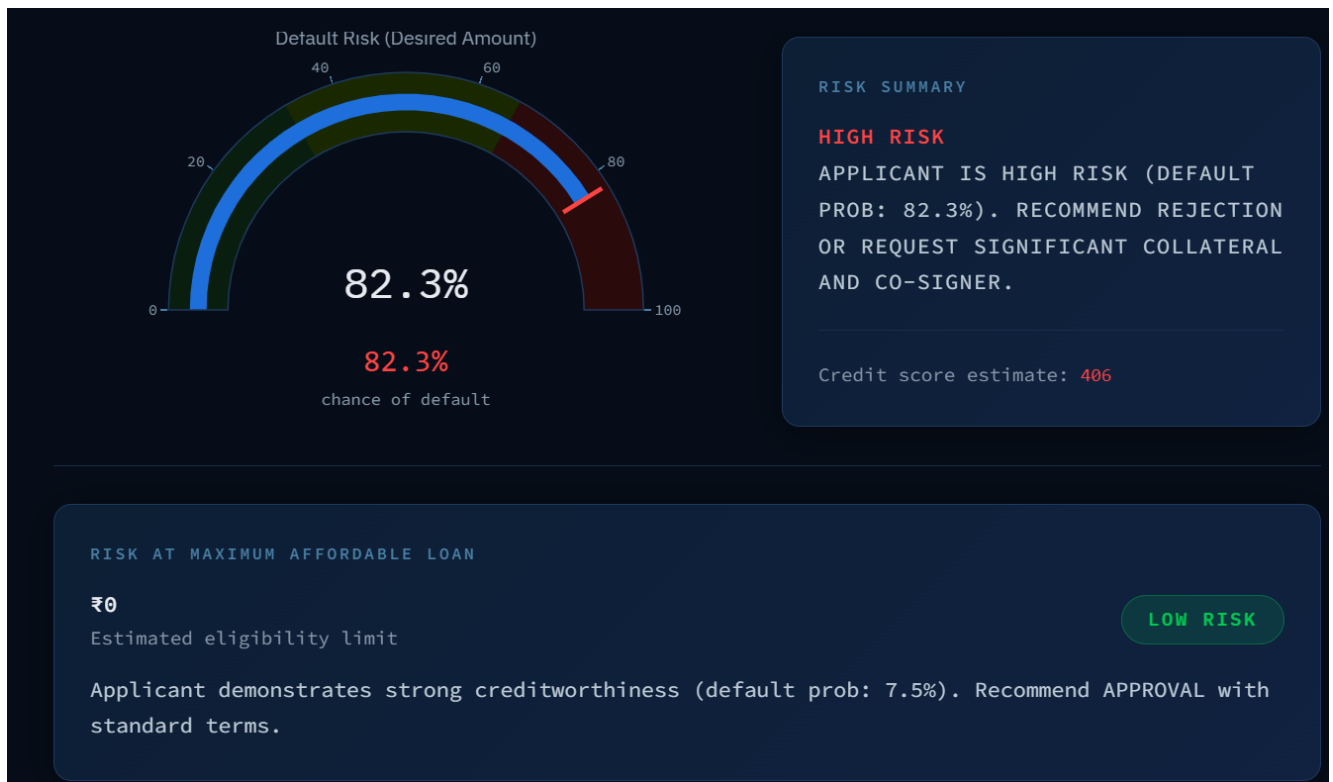


Figure 8: Risk assessment output with default probability and recommendation.

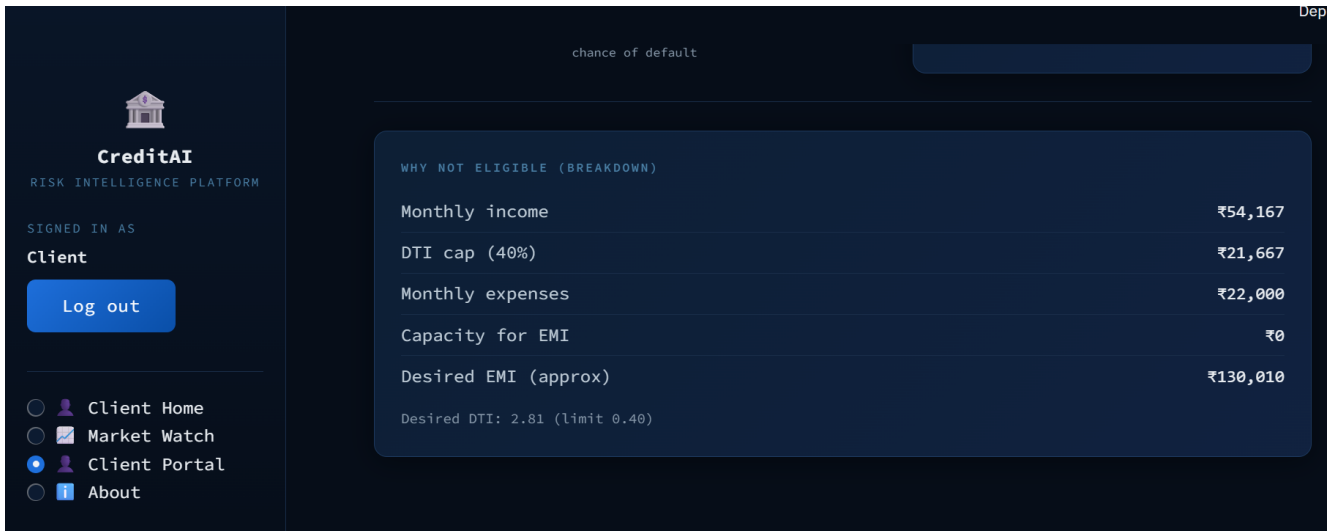


Figure 9: Eligibility breakdown showing income, DTI, and EMI capacity.

7. Discussion

The platform delivers explainable, fast credit insights while supporting both institutional evaluation and individual eligibility checks. Feature importance and risk-tier explanations improve transparency, and the integration of live assets introduces dynamic signals that are often missing from static scoring systems. The approach also enables a smoother customer experience through a client portal, while retaining rigorous evaluation controls in the bank dashboard.

8. Conclusion and Future Scope

CreditAI demonstrates a practical, deployable AI-driven credit risk solution that blends machine learning, explainability, and live financial signals. Future work includes model calibration for institution-specific policies, incorporation of bureau data, and richer behavioral signals such as transactional timelines and cash-flow stability. Additional work may also include automated decision thresholds and audit-ready model cards.

References

Paithane, P. M. (2023). Random Forest Algorithm Use for Crop Recommendation. *Journal of Engineering and Technology for Industrial Applications*, 9(43), 34–41.

Additional internal project documentation and model artifacts from the CreditAI platform (2026).